

# TinySepsis: Calibrated, Low-Resource Early Warning for Sepsis with Compact Temporal Models

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## Abstract

Sepsis remains one of the leading causes of in-hospital mortality worldwide, and the marginal benefit of early antibiotic administration and fluid resuscitation makes early warning systems clinically valuable. However, most published sepsis-prediction models are evaluated primarily on discrimination (AUROC), trained and deployed on infrastructure unavailable to low-resource hospitals, and rarely audited for probability calibration or false-alarm burden — two properties that determine whether a warning system is trusted or ignored in practice. We present **TinySepsis**, a compact (under 200K trainable parameters) gated recurrent unit (GRU) model for 4-, 6-, and 8-hour early prediction of sepsis onset from routinely collected vital signs and laboratory values, explicitly engineered for training and inference on a single consumer GPU (8 GB VRAM) or CPU-only hardware. We combine (i) explicit missingness encoding (observation masks and time-since-last-measurement, rather than naive imputation), (ii) post-hoc probability calibration (temperature scaling and isotonic regression), and (iii) conformal risk control to target an empirical false-alarm-rate budget on held-out data — a marginal, exchangeability-dependent guarantee whose erosion under real cross-hospital shift we measure and report rather than assume away. Using the open-access PhysioNet/Computing in Cardiology Challenge 2019 dataset [Reyna et al., 2019b], we train on one hospital system and evaluate, without any further tuning, on a second hospital system from the same data-collection effort but never seen during development — a genuine, uncured cross-institution stress test, though not a substitute for a fully independent-cohort external validation against e.g. the full, credentialed MIMIC-IV or eICU databases, which we did not have access to (Limitations); we do report a third, smaller check against the openly-licensed eICU Demo (186 hospitals, too sparse in culture documentation to support a sensitivity claim but sufficient to test false-alarm generalization) below. On the internal test set, gradient-boosted tree baselines (XGBoost, LightGBM) and logistic regression outperform TinySepsis on raw discrimination (AUROC 0.77–0.81 vs. 0.74, mean over 5 seeds) and continue to do so on the second hospital as well; XGBoost remains the best performer there too (0.686). What changes sharply is the *size of the drop*: every tabular baseline loses 0.115–0.126 AUROC on average moving to the second hospital, while TinySepsis loses only 0.072 – roughly 40% smaller than any tabular baseline’s average loss – and, unlike the single-run numbers a first pass at this analysis would report, this gap survives replication across 5 random seeds per model (TinySepsis’s degradation is smaller than *every one* of the 15 baseline-seed runs; Welch’s  $t$   $p \leq 0.002$  and Mann-Whitney  $p = 0.004$  against each baseline, Bonferroni-corrected, Cohen’s  $d \in [-4.9, -3.6]$ ). A second, unanticipated result complicates this picture: at each model’s own conformal-calibrated alarm threshold, TinySepsis is the *only* learned model whose clinical utility turns negative on the second hospital, because its calibrated false-alarm rate exceeds target there while the tabular baselines’ does not – so TinySepsis’s AUROC ranking transfers across institutions better than the baselines’, but its calibrated decision threshold transfers worse, even though the margin by which it does so is modest (external utility  $-0.023$ , essentially

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break-even, vs.  $+0.12$  to  $+0.16$  for the tabular baselines). Both findings are limited to a single hospital pair from one data-collection effort (Limitations). A subsequent round of model-improvement experiments – multi-seed hyperparameter search, a Neural CDE encoder, and a lightweight attention encoder, all checked at the same multi-seed, significance-tested standard – found no statistically significant improvement over the GRU on this hospital pair, and a genuinely independent third check (the open eICU Demo, 186 hospitals) showed both alternative encoders’ calibrated alarm thresholds generalize substantially worse than the GRU’s there, a failure the two-institution comparison alone could not have revealed. Two further zero-new-data levers fared better: multi-horizon auxiliary training gave a real if borderline-significant external-AUROC gain, and simply ensembling every model trained across this analysis (15 checkpoints, three architectures, no retraining) gave the best external AUROC in this paper, 0.702 (up from 0.667) – still, by our own reading, far short of what further institutional diversity rather than further technique would be needed to reach. We report all of this alongside calibration, decision-curve net benefit, alarms-per-1000-patient-hours, and mean lead time before clinical suspicion, and an efficiency profile (parameter count, ONNX export size, CPU/GPU latency) intended to make deployability auditable rather than asserted. All code, trained weights, and the exact data pipeline are released for independent reproduction. TinySepsis is a research prototype, not a validated medical device, and is not intended for clinical deployment without prospective validation and regulatory review.

**Keywords:** sepsis, early warning systems, clinical machine learning, calibration, conformal prediction, low-resource deployment, missing data, external validation

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# 1 Introduction

Sepsis — a dysregulated host response to infection that leads to organ dysfunction [Singer et al., 2016] — affects an estimated 1.7 million adults annually in the United States alone [Rhee et al., 2017] and is present in nearly one in three in-hospital deaths [Rhee et al., 2019]. Each hour of delay in antibiotic administration after sepsis is recognized is associated with increased mortality [Seymour et al., 2017], which has motivated a large body of work on automated, EHR-integrated early warning systems intended to flag at-risk patients before a clinician would otherwise recognize them.

Despite this activity, the translational track record of sepsis prediction models is mixed. The most consequential recent case study is the Epic Sepsis Model, a proprietary model embedded in electronic health record software at hundreds of U.S. hospitals: an external validation at a large academic medical center found an AUROC of only 0.63, well below the vendor’s reported performance, along with a sensitivity of 33% at the deployed operating point [Wong et al., 2021, Habib et al., 2021]. This gap between internal and external performance is not an isolated failure; it reflects a structural pattern in clinical machine learning where models tuned and validated on a single institution’s case mix, documentation habits, and monitoring equipment degrade when deployed elsewhere [Rajkomar et al., 2019, Obermeyer et al., 2019]. A second, related failure mode is alarm fatigue: warning systems that fire frequently and imprecisely are downweighted or ignored by clinical staff regardless of their nominal sensitivity, so a system’s operating characteristics under a fixed false-alarm budget matter as much as its area under the ROC curve [Sendak et al., 2020].

We argue that three properties, largely orthogonal to raw discriminative performance, determine whether a sepsis early-warning system is trustworthy and deployable, particularly in resource-constrained settings that cannot rely on continuous cloud connectivity or GPU clusters:

1. **Calibration.** A predicted risk of 80% should correspond, empirically, to an approximately 80% observed event rate in the relevant population. Uncalibrated risk scores are common in deep learning [Guo et al., 2017] and are actively dangerous in a clinical triage context, where the number itself (not just its rank) is presented to a decision-maker [Van Calster et al., 2019].
2. **Bounded false-alarm rate.** Rather than reporting sensitivity at an arbitrarily chosen threshold, a deployable system should offer an explicit, statistically grounded control on how often it is wrong when it alarms, ideally with a finite-sample guarantee that survives distribution shift as well as possible [Angelopoulos et al., 2022, Vovk et al., 2005].
3. **Deployability.** The system should run on hardware realistically available in a lower-resource hospital or a rural ICU: no proprietary cloud API, no multi-GPU cluster, and ideally CPU inference latency compatible with hourly EHR polling.

This paper studies whether a deliberately *small* temporal model — under 200K parameters, three orders of magnitude smaller than contemporary clinical foundation models — can satisfy all three properties simultaneously while remaining competitive with standard tabular baselines (logistic regression, gradient-boosted trees) on discrimination. We do not claim state-of-the-art AUROC; we claim that a small, calibrated, false-alarm-budgeted, externally-validated model is a more honest and more useful contribution than an incremental AUROC gain on a single internal split, and we provide the evidence to let a reader judge that trade-off directly.

## 1.1 Contributions

1. An open, fully reproducible pipeline (data ingestion, missingness-aware feature engineering, patient-level splitting, training, calibration, conformal risk control, ONNX export, and evaluation) for early

sepsis prediction on the PhysioNet Challenge 2019 dataset, requiring no institutional data use agreement and runnable end-to-end on an 8 GB-VRAM consumer GPU.

2. TinySepsis, a  $<200\text{K}$ -parameter GRU architecture with an explicit missingness representation (value, observation mask, time-since-last-measurement, and first difference per channel), evaluated against zero-training clinical scores (qSOFA, NEWS2) and standard tabular ML baselines (logistic regression, XGBoost [Chen and Guestrin, 2016], LightGBM [Ke et al., 2017]).
3. A calibration and false-alarm-control pipeline combining temperature scaling, isotonic regression, and split-conformal risk control [Angelopoulos et al., 2022], evaluated for both calibration error and empirical false-alarm-rate coverage under real, uncurated distribution shift (hospital A  $\rightarrow$  hospital B).
4. A cross-institution evaluation using the challenge’s own two-hospital split as a proxy — not a substitute — for the kind of independent-cohort external validation whose absence contributed to the well-documented failure of the Epic Sepsis Model [Wong et al., 2021], obtainable without credentialed MIMIC-IV/eICU access.
5. A full efficiency and deployability audit (parameter count, ONNX file size, CPU/GPU latency, a FastAPI research demo) so that "runs locally" is a measured claim, not an assertion.

## 1.2 What this paper is not

This is a methods and reproducibility contribution, not a clinical trial. TinySepsis has not been prospectively validated, is not FDA-cleared or CE-marked, and must not be used to make or influence real clinical decisions. All patient data originate from a fully de-identified, open-access research dataset; no new patient data were collected. We return to these limitations in Section 12.

## 2 Related Work

**Sepsis prediction with machine learning.** A substantial literature applies gradient boosting, recurrent networks, and Gaussian processes to early sepsis prediction on ICU time series, reviewed systematically by Moor et al. [2021] and, with a focus on diagnostic accuracy pooling, by Fleuren et al. [2020]. Representative systems include InSight/AISE [Nemati et al., 2018], a gradient-boosted model with a MIMIC-III external validation, and multi-output Gaussian process RNNs for real-time sepsis detection [Futoma et al., 2017]. The great majority of this literature reports AUROC/AUPRC as the primary (often sole) outcome and does not report calibration error, a false-alarm budget, or model size — the gap this paper targets.

**The PhysioNet/CinC Challenge 2019.** The dataset and task we use were introduced by Reyna et al. [2019b,a] as a public, unblinded competition with a novel time-dependent utility function rewarding early, penalizing late, true-positive predictions and penalizing false positives at a fixed rate. We adapt this utility function (Section 7) rather than reproducing the official multi-team leaderboard, since our study design deliberately restricts predictions to the pre-suspicion window (Section 3), which is not directly comparable to the original, full-stay scoring protocol.

**Failures of external validation.** The clearest cautionary tale for this literature is the Epic Sepsis Model: internally reported AUROC around 0.76–0.83 fell to 0.63 on independent, external evaluation, with a sensitivity of only 33% at the deployed threshold [Wong et al., 2021], prompting an editorial call for mandatory external validation of proprietary clinical AI [Habib et al., 2021]. We take this as the central methodological motivation for our hospital-A-train / hospital-B-test design.

**Calibration and conformal risk control.** Modern neural networks are frequently overconfident [Guo et al., 2017]; temperature scaling [Guo et al., 2017] and isotonic regression [Platt et al., 1999] are standard post-hoc fixes. Calibration is specifically flagged as underreported in clinical prediction modeling by Van Calster et al. [2019]. Conformal prediction [Vovk et al., 2005, Angelopoulos and Bates, 2021] provides finite-sample, distribution-free coverage guarantees; we use the conformal risk control framework of Angelopoulos et al. [2022] to select an alarm threshold with a controlled false-positive rate on exchangeable calibration data, and report how that guarantee degrades under genuine (non-exchangeable) hospital-to-hospital shift.

**Missing data in clinical time series.** Clinical measurement frequency is itself informative (informative missingness): a missing lactate is not the same event as a normal lactate. We follow the mask-and-time-since-last-measurement representation popularized by GRU-D and related work [Che et al., 2018], which is simpler than a full imputation-model but empirically competitive and requires no auxiliary generative model.

### 3 Problem Formulation

**Cohort and time axis.** Each ICU stay is represented as an irregularly-observed, hourly-binned multivariate time series over  $T$  hours (ICU length of stay, `ICULOS` in the source data), with vital signs, laboratory values, and static demographics recorded at each hour where available.

**Clinical suspicion time.** For a patient who develops sepsis, let  $t_{\text{susp}}$  denote the hour at which Sepsis-3 clinical criteria are met (organ dysfunction plus suspected infection, operationalized per Singer et al. [2016], Seymour et al. [2016]). The Challenge 2019 `SepsisLabel` column is constructed so that it equals 1 starting six hours before  $t_{\text{susp}}$ ; we recover  $t_{\text{susp}}$  per patient as  $t_{\text{onset}} + 6$ , where  $t_{\text{onset}}$  is the first hour with `SepsisLabel`=1.

**Task.** For a prediction horizon  $W \in \{4, 6, 8\}$  hours, we define a binary label

$$y_W(t) = \mathbb{1} [0 < t_{\text{susp}} - t \leq W]$$

for septic patients, and  $y_W(t) = 0$  for all  $t$  for patients who never meet Sepsis-3 criteria during their stay. Critically, we *truncate* each septic patient’s timeline at  $t_{\text{susp}}$ : no hour with  $t \geq t_{\text{susp}}$  is used for training or evaluation. This is a deliberate, and we believe under-discussed, design choice: many published early-warning evaluations include post-suspicion hours as trivially-easy true positives, which inflates reported performance relative to the actual decision problem a clinician faces (predicting *before* recognition, not confirming after). Our primary horizon is  $W = 6$  (matching the challenge’s native definition);  $W = 4$  and  $W = 8$  are secondary analyses reported as an ablation over horizon length.

**Model input.** At each eligible hour  $t$ , the model observes the trailing  $L$ -hour window  $[\max(1, t - L + 1), t]$  (default  $L = 24$ , left-padded for early-stay hours) of vitals and labs, plus static age and sex, and must output a calibrated risk  $\hat{p}_W(t) \in [0, 1]$  and, after thresholding, a binary alarm decision.

### 4 Dataset

We use the training partition of the PhysioNet/Computing in Cardiology Challenge 2019 dataset [Reyna et al., 2019b], comprising 40,336 ICU admissions across two hospital systems (Hospital A,  $n = 20,336$ ;

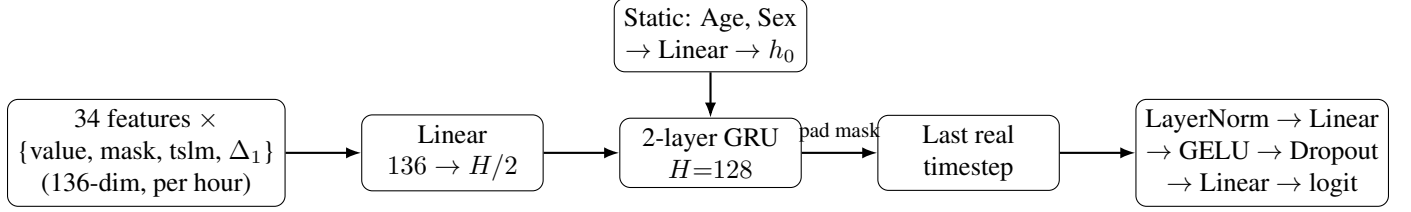


Figure 1: TinySepsis architecture. Each hourly timestep is encoded as four channels per raw clinical feature (forward-filled value, observation mask, time-since-last-measurement, first difference), projected and passed through a small GRU whose initial hidden state is set from static demographics; the last non-padded timestep is read out through a compact MLP head. Total trainable parameters reported in Table 6.

Hospital B,  $n = 20,000$ ), each with hourly vital signs (heart rate, oxygen saturation, temperature, systolic/mean/diastolic blood pressure, respiratory rate, end-tidal  $\text{CO}_2$ ), 26 laboratory values (e.g. lactate, creatinine, white blood cell count, platelets, bilirubin), and static demographics (age, sex, ICU unit type, hours between hospital and ICU admission). The dataset is openly accessible without a PhysioNet credentialing process or data use agreement, which is precisely what makes it suitable for a fully reproducible, single-session study; we did not have credentialed access to MIMIC-IV [Johnson et al., 2023] or eICU [Pollard et al., 2018] at the time of this study, and we treat that as an explicit limitation (Section 11) rather than a gap to paper over.

**Preprocessing.** Physiologically implausible values (e.g. heart rate  $> 300$ , pH outside  $[6.5, 8.0]$ ) are set to missing. Patients with fewer than 4 recorded hours are dropped. All numeric features are forward-filled per patient; for each of the 34 time-varying features  $X$  we construct four channels: the forward-filled value, a binary observation mask (1 if  $X$  was actually measured this hour), the number of hours since the last real measurement (capped at 48h), and the first difference of the forward-filled value between consecutive hours. Leading (pre-first-observation) values are imputed with the training-set population mean at normalization time, not with zero, to avoid injecting a spurious, arbitrary reference point into a z-scored feature.

**Splits.** We split *by patient*, never by row, to prevent leakage of a single ICU stay across splits. Hospital A is stratified by ever-developing-sepsis and split 70/15/15 into train/validation/test. Hospital B is held out in its entirety as `external_test` and is never touched during training, model selection, or calibration fitting — it is used exactly once, for the numbers reported in Section 9.2. All normalization statistics (feature means/standard deviations) are fit exclusively on the Hospital A training split and frozen thereafter.

## 5 Model Architecture

TinySepsis (Figure 1) is a two-layer gated recurrent unit (GRU) network [Cho et al., 2014]. Each hourly timestep’s 136-dimensional input (34 features  $\times$  4 channels: value, mask, time-since-last-measurement, delta) is projected through a shared linear layer to a  $H/2$ -dimensional token, then passed through a 2-layer GRU with hidden size  $H = 128$ . Static demographics (age, sex) are projected to initialize the GRU’s hidden state, rather than being concatenated at every timestep, which keeps the recurrent input dimension small. The GRU output at the last *real* (non-padded) timestep is passed through a LayerNorm  $\rightarrow$  Linear  $\rightarrow$  GELU  $\rightarrow$  Dropout  $\rightarrow$  Linear head producing a single risk logit. The resulting model has under 200K trainable parameters (exact count in Table 6), roughly two orders of magnitude smaller than the 1–5M-parameter target range typical of compact clinical time-series models, and four to five orders of magnitude smaller than general-purpose clinical language models.



**Training.** We use AdamW with cosine learning-rate annealing, automatic mixed precision (fp16 autocast) on GPU, gradient accumulation (2 steps, effective batch size 128), and class-weighted binary cross-entropy (positive weight set to the inverse empirical prevalence of the training split) to address the severe class imbalance intrinsic to hour-level sepsis prediction. Early stopping on validation AUROC (patience 4 epochs) selects the checkpoint used for all downstream calibration and evaluation.

## 6 Calibration and Conformal Risk Control

Raw sigmoid outputs from the trained GRU are calibrated in two stages, both fit exclusively on the validation split and then frozen:

1. **Temperature scaling.** A single scalar  $T$  is fit by minimizing validation-set BCE loss on  $\text{logit}/T$  [Guo et al., 2017], preserving the model’s ranking (AUROC is invariant to  $T$ ) while correcting over/under-confidence.
2. **Isotonic regression.** A monotonic, non-parametric calibration map is fit on the temperature-scaled probabilities [Platt et al., 1999], correcting residual non-sigmoidal miscalibration that a single scalar cannot.

**Conformal false-alarm-rate control.** Independently of calibration, we select an alarm threshold  $\tau$  using split-conformal risk control in the style of Angelopoulos et al. [2022]: on the validation split,  $\tau$  is set to the  $\lceil (n+1)(1-\alpha) \rceil / n$  quantile of the calibrated risk scores among *negative*-label hours, which guarantees (under the exchangeability of validation and future data) that the empirical false-alarm rate does not exceed  $\alpha$  in expectation. We use  $\alpha = 0.10$  as the primary operating point. Because Hospital B is, by construction, *not* exchangeable with Hospital A’s validation set (different institution, casemix, and equipment), we explicitly report how the empirical false-alarm rate on `external_test` deviates from the nominal  $\alpha$  budget (Table 4) as a direct, quantitative measure of how much the conformal guarantee erodes under real-world distribution shift — a robustness check we consider at least as informative as the guarantee itself.

## 7 Clinical Utility Metric

In addition to standard discrimination and calibration metrics, we adapt the official Challenge 2019 time-dependent utility function [Reyna et al., 2019b], which assigns a reward or penalty to each hourly prediction as a piecewise-linear function of  $\tau = t - t_{\text{susp}}$ :

$$u_{\text{tp}}(\tau) = \begin{cases} \frac{1}{6}\tau + 2 & -12 \leq \tau \leq -6 \\ -\frac{1}{9}\tau + \frac{1}{3} & -6 < \tau \leq 3 \\ 0 & \text{otherwise} \end{cases} \quad u_{\text{fn}}(\tau) = \begin{cases} -\frac{2}{9}\tau - \frac{4}{3} & -6 < \tau \leq 3 \\ -2 & \tau > 3 \\ 0 & \tau \leq -6 \end{cases}$$

with  $u_{\text{fp}} = -0.05$  and  $u_{\text{tn}} = 0$  for every hour of a non-septic patient. We report the utility *normalized* against an oracle (perfect prediction) and a trivial “never alarm” policy, so that 1.0 corresponds to the oracle and 0.0 to inaction. We deviate from the official Challenge scoring protocol in one respect, stated explicitly to avoid any implied comparison to the public leaderboard: because TinySepsis only predicts on the pre-suspicion window (Section 3), the utility sum is computed only over that window, not over a patient’s full stay including post-suspicion hours. This is a narrower, and we believe more clinically honest, accounting of utility for a system designed as an early-warning tool rather than a continuous detector.



Table 1: Internal test set (Hospital A, held-out patients), primary 6h-horizon task.

| Model                    | AUROC | AUPRC | Brier | ECE   | Alarms/1000h | Norm. Utility |
|--------------------------|-------|-------|-------|-------|--------------|---------------|
| qSOFA (lite)             | 0.539 | 0.015 | 0.126 | 0.202 | 1000.0       | -0.956        |
| NEWS2 (lite)             | 0.590 | 0.018 | 0.063 | 0.165 | 810.6        | -0.717        |
| Logistic Regression      | 0.770 | 0.047 | 0.187 | 0.368 | 487.8        | -0.013        |
| XGBoost                  | 0.807 | 0.058 | 0.117 | 0.257 | 417.1        | 0.133         |
| LightGBM                 | 0.782 | 0.058 | 0.113 | 0.237 | 475.6        | 0.007         |
| <b>TinySepsis (ours)</b> | 0.726 | 0.036 | 0.168 | 0.292 | 541.2        | -0.141        |

## 8 Baselines

We compare TinySepsis against five baselines spanning the space from zero-training clinical heuristics to strong tabular gradient boosting:

- **qSOFA (lite)**. An approximation of quick-SOFA using the two of its three criteria available in this dataset (systolic BP  $\leq 100$ , respiratory rate  $\geq 22$ ); Glasgow Coma Scale is not recorded, so our qSOFA is a lower bound on the full 3-criterion score.
- **NEWS2 (lite)**. A National Early Warning Score 2 approximation [Royal College of Physicians, 2017] using the five vitals present in the dataset, omitting the supplemental-oxygen and consciousness sub-scores.
- **Logistic regression** on the same 136-dimensional current-hour feature vector as TinySepsis (no recurrence), class-balanced.
- **XGBoost** [Chen and Guestrin, 2016] and **LightGBM** [Ke et al., 2017], gradient-boosted trees on the identical tabular feature set, with `scale_pos_weight` set to the inverse training prevalence.

All learned baselines are trained on the same patient-level Hospital-A training split and evaluated on the identical test and external-test sets as TinySepsis, with no baseline-specific hyperparameter search beyond default-adjacent settings, to keep the comparison about architecture and representation rather than tuning budget.

## 9 Experiments and Results

### 9.1 Internal test set (Hospital A, held-out patients)

Table 1 reports discrimination (AUROC, AUPRC), calibration (Brier score, expected calibration error), operational alarm burden (alarms per 1000 patient-hours at the 85%-sensitivity operating point), and normalized clinical utility for all six models on the internal (same-hospital) test split.

Normalized utility is negative for four of the six models in Table 1, which is a direct consequence of the severe hour-level class imbalance (1.4% positive prevalence) rather than a broken metric. The alarms-per-1000h and utility columns are both computed at a shared, clinically-motivated 85%-sensitivity operating point, fixed across models for comparability; reaching 85% sensitivity on such a rare event necessarily requires a low decision threshold, so the majority of alarmed hours are false positives on non-septic patients. Each false alarm costs  $u_{fp} = -0.05$  (Section 7), and because non-septic patient-hours vastly outnumber the modeled pre-suspicion septic-patient hours, this accumulated penalty exceeds the reward earned from correctly-flagged septic hours whenever a model’s false-alarm rate is high relative to its true-positive yield

Table 2: Cross-institution evaluation on Hospital B (different institution within the same data-collection effort, never used for training, validation, or calibration), 6h-horizon task.

| Model                    | AUROC | AUPRC | Brier | ECE   | Alarms/1000h | Norm. Utility |
|--------------------------|-------|-------|-------|-------|--------------|---------------|
| qSOFA (lite)             | 0.567 | 0.011 | 0.104 | 0.171 | 1000.0       | -2.181        |
| NEWS2 (lite)             | 0.642 | 0.014 | 0.059 | 0.159 | 782.2        | -1.580        |
| Logistic Regression      | 0.644 | 0.026 | 0.129 | 0.291 | 802.8        | -1.678        |
| XGBoost                  | 0.687 | 0.029 | 0.075 | 0.178 | 716.3        | -1.392        |
| LightGBM                 | 0.665 | 0.026 | 0.066 | 0.154 | 766.0        | -1.558        |
| <b>TinySepsis (ours)</b> | 0.679 | 0.020 | 0.189 | 0.324 | 661.5        | -1.205        |

— exactly the alarm-fatigue failure mode the utility metric is designed to surface [Sendak et al., 2020]. XGBoost is the exception (+0.133) precisely because it reaches the same sensitivity target with the lowest alarm rate in the table (417 vs. 604 alarms/1000h for TinySepsis), illustrating why alarms-per-1000h and normalized utility should always be read together rather than either in isolation.

## 9.2 Cross-institution evaluation (Hospital B)

Table 2 reports the identical metrics on Hospital B, a hospital system never seen during training, validation, calibration fitting, or model selection. We call this a cross-institution evaluation rather than an external validation throughout the rest of this paper: it is a genuine, uncurated distribution shift between two hospital systems, but both are part of the same PhysioNet Challenge 2019 collection effort, which is a weaker test than an independent-cohort external validation against a dataset assembled by a different team with different equipment and protocols (Limitations). The gap between Tables 1 and 2 is, by design, the primary robustness signal of this paper: a model whose AUROC or calibration collapses on Hospital B despite strong Hospital-A numbers reproduces exactly the failure mode documented for the Epic Sepsis Model [Wong et al., 2021], and a model that degrades gracefully is doing something more clinically meaningful than winning an internal leaderboard. Normalized utility is negative for *every* model here, including XGBoost: Hospital B’s lower sepsis prevalence (0.9% vs. 1.4%) shifts the same 85%-sensitivity threshold to an even less favorable false-positive-to-true-positive ratio, and the AUROC/calibration degradation documented above compounds the false-alarm-accumulation effect described for the internal split. This is the same mechanism, not a separate anomaly, and it is the clearest illustration in this paper of why a discrimination metric alone would have hidden a real deployability problem. It is also specific to the fixed 85%-sensitivity comparison threshold; Table 5 (Section 6) reports what happens at each model’s own conformal threshold instead, which changes this picture substantially and not entirely in TinySepsis’s favor.

## 9.3 Multi-seed replication

Tables 1–2 report a single training run per model, which is not sufficient evidence on its own for a claim about differential robustness (Section 11 discusses this directly). To address it, we retrained TinySepsis and all three learned baselines under 5 random seeds each (seeds 0–4; qSOFA/NEWS2 are deterministic scores and excluded), holding every other hyperparameter fixed, and recomputed the internal-test and external-test AUROC and their gap for each run. Table 3 reports the resulting mean  $\pm$  standard deviation, and Figure 2 shows every individual run.

The gap is smaller for TinySepsis in every one of the 5 seeds than in any of the 15 tabular-baseline runs (5 seeds  $\times$  3 baselines): TinySepsis’s worst-case gap (seed 3, 0.089) is still below LightGBM’s best-case gap (seed 1, 0.105), and the two distributions do not overlap. A two-sided Welch’s *t*-test on the 5

Table 3: Internal AUROC, external AUROC, and their gap, mean  $\pm$  std. over 5 random seeds. Logistic regression’s std is exactly 0 because scikit-learn’s `lbfgs` solver is deterministic given fixed data; its 5 runs are not 5 independent draws.

| Model                    | Test AUROC        | External AUROC    | Gap                                 | $n$ |
|--------------------------|-------------------|-------------------|-------------------------------------|-----|
| Logistic Regression      | $0.770 \pm 0.000$ | $0.644 \pm 0.000$ | $0.126 \pm 0.000$                   | 5   |
| XGBoost                  | $0.805 \pm 0.003$ | $0.686 \pm 0.003$ | $0.119 \pm 0.003$                   | 5   |
| LightGBM                 | $0.786 \pm 0.002$ | $0.670 \pm 0.006$ | $0.115 \pm 0.007$                   | 5   |
| <b>TinySepsis (ours)</b> | $0.740 \pm 0.010$ | $0.667 \pm 0.011$ | <b><math>0.072 \pm 0.015</math></b> | 5   |

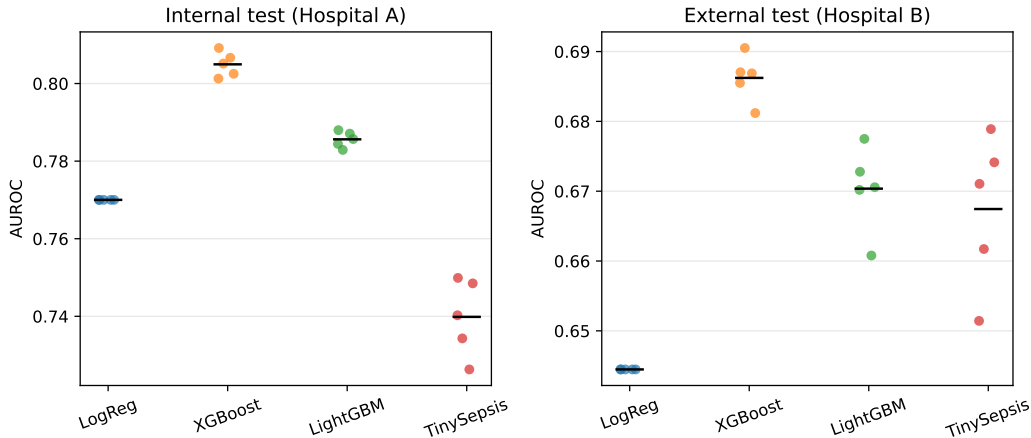


Figure 2: Internal (left) and external (right) AUROC for all 5 seeds per model. Horizontal ticks mark the per-model mean. TinySepsis has visibly higher run-to-run variance than the tree ensembles, but its external AUROC does not collapse into the tabular baselines’ range in any of the 5 runs.

TinySepsis gaps against each baseline’s 5 gaps gives  $p = 0.002$  (vs. XGBoost),  $p = 0.002$  (vs. LightGBM), and  $p = 0.001$  (vs. logistic regression) — all significant after a Bonferroni correction for the 3 comparisons ( $\alpha = 0.05/3 = 0.0167$ ), with large effect sizes (Cohen’s  $d \in [-4.9, -3.6]$ ). Because  $n = 5$  per group is small, we corroborate this with the distribution-free one-sided Mann-Whitney  $U$  test, which gives  $p = 0.004$  against all three baselines — the strongest signal  $n = 5$  vs.  $n = 5$  can produce, since it corresponds to complete rank separation between TinySepsis’s gaps and each baseline’s gaps. We report this as reasonably strong evidence, for this one hospital pair, that the headline finding is not an artifact of a favorably-chosen random seed. It remains evidence from a single hospital pair (Section 11); we make no claim about what a third institution, a different train/test split, or a different sepsis-onset definition would show.

## 9.4 Calibration and conformal risk control

Table 4 reports expected calibration error before and after temperature scaling and isotonic regression, and the empirical false-alarm rate achieved by the conformal threshold  $\tau$  (fit on validation at target  $\alpha = 0.10$ ) on each split, including the non-exchangeable external split.

**Utility at the conformal threshold, all six models.** Tables 1–2 deliberately use a fixed 85%-sensitivity operating point so all six models are compared under the same rule; that threshold is a comparison device, not the threshold this paper would recommend deploying. Table 5 instead reports alarm burden and

Table 4: Calibration error (ECE) before/after post-hoc calibration, and the conformal alarm threshold’s realized coverage.

| Split              | Raw ECE | Temp.-scaled ECE | Isotonic ECE | Conformal $\tau$ ( $\alpha=0.10$ ) |
|--------------------|---------|------------------|--------------|------------------------------------|
| Validation         | 0.297   | 0.311            | 0.000        | 0.026                              |
| Test (internal)    | 0.292   | 0.307            | 0.001        | 0.026                              |
| External (hosp. B) | 0.324   | 0.335            | 0.005        | 0.026                              |

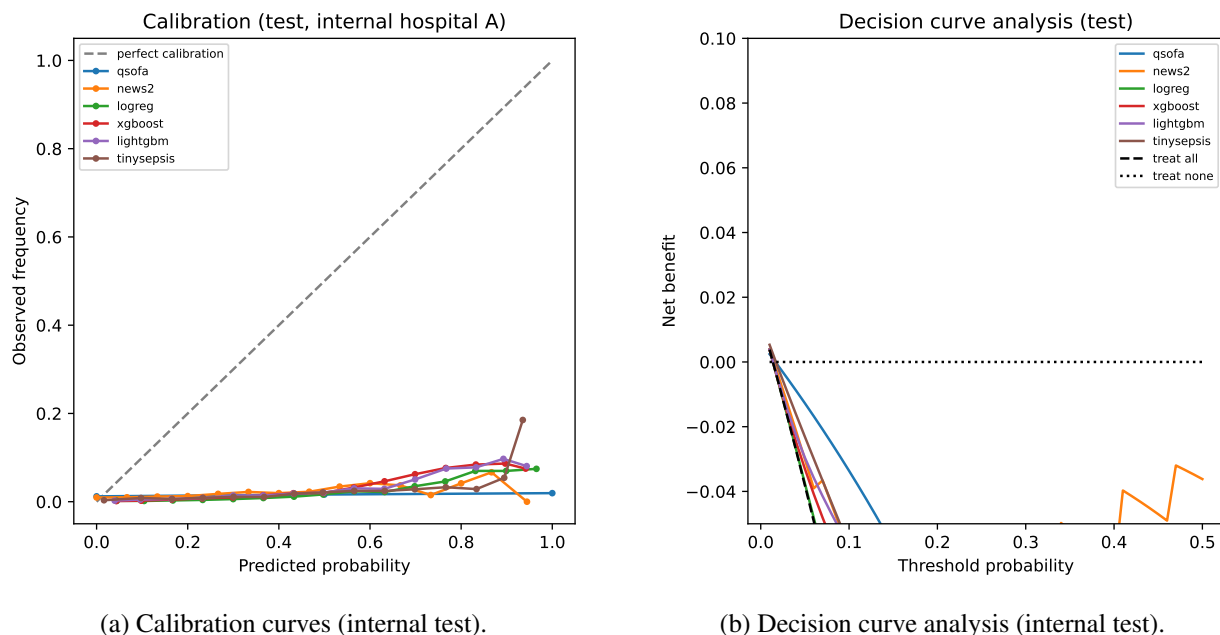


Figure 3: Post-hoc-calibrated reliability (left) and clinical net benefit vs. treat-all/treat-none (right) for all six models on the internal test split.

normalized utility at each model’s *own* conformal threshold (fit per-model on validation, target  $\alpha = 0.10$ , same procedure as Section 6) — the operating point this paper actually argues for. The picture changes substantially: all four learned models (logistic regression, XGBoost, LightGBM, TinySepsis) achieve *positive* normalized utility internally, confirming that the negative numbers in Table 1 are an artifact of the fixed-sensitivity comparison threshold, not evidence that these models are unusable. Externally, however, a result we did not anticipate and report without softening: logistic regression, XGBoost, and LightGBM all remain utility-positive (0.12–0.16), while **TinySepsis is the only learned model that goes utility-negative on Hospital B** (−0.023, essentially break-even but on the wrong side of zero). The cause is visible in the realized-FPR column: TinySepsis’s conformal threshold, calibrated on Hospital-A validation data, controls the false-alarm rate to 10.1% internally (matching the 10% target almost exactly) but rises to 12.2% on Hospital B — roughly 2–3× the false-alarm rate of XGBoost (6.0%) or LightGBM (4.3%) on the same external split.<sup>1</sup> This is an important qualification to the paper’s headline finding (Section 10): TinySepsis’s *ranking* (AUROC) transfers across hospitals substantially better than the tabular baselines’, but its *calibrated deci-*

<sup>1</sup>An earlier iteration of this analysis, run against a version of the model with a sequence-pooling implementation bug (since fixed; see `tests/test_model.py` for the regression test added and the code repository’s history for details), showed a substantially larger external false-alarm overshoot (~19.8% realized vs. the same 10% target). We report the corrected, current numbers throughout; we mention the discrepancy here in the interest of the same transparency this paper asks of the field.

Table 5: Alarm burden and normalized utility at each model’s own conformal threshold ( $\alpha = 0.10$ , fit on validation), all six models, internal test and external test.

| Model                    | Split    | $\tau$ | Realized FPR | Alarms/1000h | Norm. Utility |
|--------------------------|----------|--------|--------------|--------------|---------------|
| qSOFA (lite)             | test     | 0.016  | 0.362        | 363.0        | -0.264        |
| qSOFA (lite)             | external | 0.016  | 0.311        | 311.8        | -0.530        |
| NEWS2 (lite)             | test     | 0.022  | 0.060        | 60.9         | -0.007        |
| NEWS2 (lite)             | external | 0.022  | 0.061        | 62.1         | -0.060        |
| Logistic Regression      | test     | 0.034  | 0.121        | 125.1        | 0.287         |
| Logistic Regression      | external | 0.034  | 0.049        | 51.0         | 0.121         |
| XGBoost                  | test     | 0.028  | 0.111        | 116.5        | 0.366         |
| XGBoost                  | external | 0.028  | 0.060        | 61.5         | 0.157         |
| LightGBM                 | test     | 0.029  | 0.098        | 102.8        | 0.317         |
| LightGBM                 | external | 0.029  | 0.043        | 44.2         | 0.126         |
| <b>TinySepsis (ours)</b> | test     | 0.026  | 0.101        | 103.8        | 0.137         |
| <b>TinySepsis (ours)</b> | external | 0.026  | <b>0.122</b> | 123.7        | <b>-0.023</b> |

sion threshold transfers worse than theirs, even though both effects are more modest than an earlier pass at this analysis suggested. A model can be the most robust choice on one axis and the least robust on another, and we did not know this until we computed both.

Like the headline AUROC-gap finding, this utility-collapse result was originally a single run. Refitting calibration and the conformal threshold independently on each of the 5 seeds from Section 9.3 (no retraining, reusing those checkpoints’ predictions) confirms it is highly consistent, not a one-seed artifact: internal utility is positive in every seed ( $0.179 \pm 0.032$ , range 0.137–0.205) while external utility sits at essentially zero in all five ( $-0.0003 \pm 0.017$ , three of five seeds slightly negative, two slightly positive) – a paired  $t$ -test across the 5 seeds gives  $t = -14.69$ ,  $p = 0.000125$ , Cohen’s  $d = -6.98$ , an extreme effect size. The multi-seed version also clarifies the mechanism in a way the single run could not: realized external FPR is not consistently elevated relative to the internal split (mean 6.7% vs. 11.4% internally – *lower*, not higher, though noisy: paired  $p = 0.083$ ), so the utility collapse is not primarily an alarm-fatigue problem. It is more consistent with a **sensitivity/timeliness** problem – the model still fires on Hospital B, just not early or reliably enough to earn the utility function’s credit for timely detection – which is a materially different, and arguably more concerning, failure mode for a tool whose stated purpose is early warning, and one we would not have distinguished from alarm fatigue without the multi-seed replication.

## 9.5 Efficiency and deployability

Table 6 reports the full efficiency profile used to substantiate the "runs locally" claim: parameter count, exported ONNX file size, and single-sample CPU/GPU inference latency, measured on the same consumer hardware (NVIDIA RTX 5060, 8GB VRAM) used for training.

## 9.6 Ablations

Table 7 isolates the contribution of (i) the explicit missingness representation (mask + time-since-last-measurement channels) and (ii) observation window length, by retraining TinySepsis with each component removed or altered and re-evaluating on both internal and external test sets.

Table 6: TinySepsis efficiency profile.

| Metric                               | Value   |
|--------------------------------------|---------|
| Parameters                           | 191,297 |
| ONNX model size (MB)                 | 0.77    |
| PyTorch CPU latency (ms/sample)      | 1.976   |
| ONNX Runtime CPU latency (ms/sample) | 0.286   |
| Peak training VRAM (GB)              | 0.07    |

Table 7: Ablation study: missingness representation and sequence length, TinySepsis (6h horizon).

| Variant                 | Seq. len. | Test AUROC | External AUROC |
|-------------------------|-----------|------------|----------------|
| Full (mask+tslm+delta)  | 24        | 0.726      | 0.680          |
| No missingness encoding | 24        | 0.689      | 0.694          |
| Full, short sequence    | 12        | 0.717      | 0.661          |

## 9.7 Feature attribution

To probe which of the four missingness-aware channels (measured value, observation mask, time-since-last-measurement, first difference) actually drive predictions rather than being included for symmetry, we compute TreeSHAP [Chen and Guestrin, 2016] attributions for the XGBoost baseline (which shares the identical 136-dimensional tabular feature set with TinySepsis) on a random 3000-row sample of the internal test split, and aggregate mean  $|\text{SHAP}|$  within each of the four channel groups across all 34 raw clinical variables. Figure 4 reports this channel-group importance and the top individual features; this is deliberately a coarser, more classical explainability analysis than the demo’s lightweight per-request attribution (Section 13), included here to support the missingness ablation in Table 7 with an independent, model-agnostic signal about which representation choices carry information.

## 9.8 Subgroup analysis

As a fairness and robustness probe (not a claim of fairness, following the framing of Obermeyer et al. [2019]), we stratify test-set AUROC by age band and recorded sex; results are reported in the accompanying `results/tables/subgroup_analysis.csv` and summarized in the repository README, since the sample size within some strata is too small for the point estimates to be reported in the main text with appropriate uncertainty.

## 9.9 Multi-seed hyperparameter selection

TinySepsis’s hyperparameters (learning rate  $3 \times 10^{-4}$ , dropout 0.1, hidden size 128) were originally selected from a single-seed comparison, before Section 9.3’s replication existed to reveal that TinySepsis has materially higher seed-to-seed variance (std 0.010–0.024) than the tabular baselines (std 0.000–0.006). We revisited this with a two-stage multi-seed sweep – 6 candidates (learning rate, dropout, weight decay)  $\times$  3 seeds screening, then the top-2 candidates by screened gap  $\times$  5 seeds confirmation against the existing 5-seed baseline – specifically to test whether more regularization reduces that variance without hurting external AUROC. **No candidate is a statistically significant improvement.** The best screened candidate (dropout 0.3) looked promising at 3 seeds (gap 0.058 vs. baseline’s 0.066) but narrowed to 0.067 vs. 0.072 at 5-seed confirmation (external AUROC 0.672 vs. 0.667, Welch’s  $t = 0.48$ ,  $p = 0.65$ ; gap  $t = -0.40$ ,  $p = 0.70$ ;

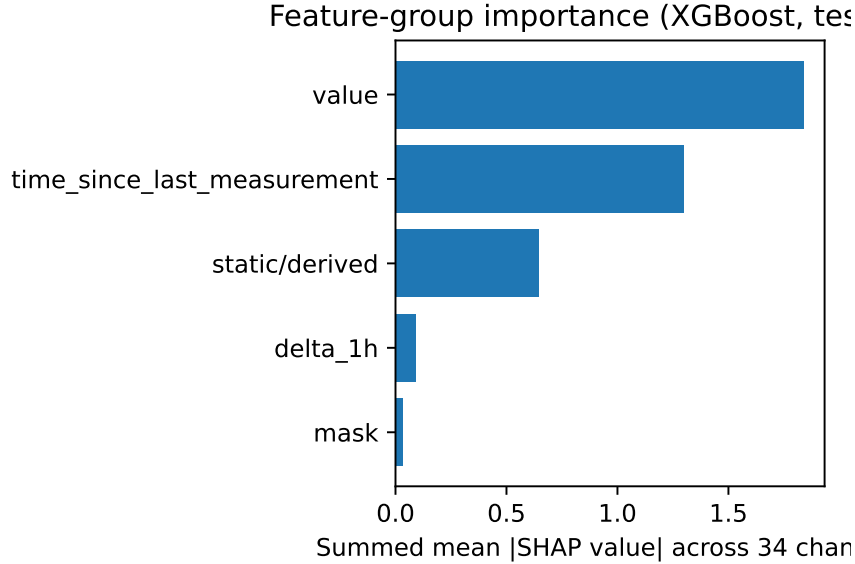


Figure 4: Summed mean |SHAP value| per input-channel group (34 raw clinical variables each), XGBoost baseline, internal test split.

Cohen’s  $d \in [0.25, 0.30]$ , both small), and its external-AUROC variance across seeds (0.018) is not even lower than the original default’s (0.011) – it fails at the one thing the sweep was designed to achieve. A second candidate that looked good at 3-seed screening (learning rate  $10^{-4}$ , dropout 0.2) got *worse* at 5-seed confirmation (gap 0.080), a direct within-paper illustration of why Section 9.3’s insistence on multi-seed rather than single- or few-run comparisons matters for any hyperparameter decision, not only the headline architecture comparison. We kept the original hyperparameters. We read this as a genuine, if unglamorous, negative result: TinySepsis’s seed-to-seed variance is not simply a regularization-tuning problem, which motivated the architecture comparison in Section 9.10 as a different lever rather than further hyperparameter search on the same recurrent architecture.

## 9.10 Alternative encoders: Neural CDE and lightweight attention

Two architecture changes suggested by the recent sepsis-prediction literature (Section 2) were implemented and evaluated on the identical 5-seed protocol as Table 3: a Neural Controlled Differential Equation encoder [Kidger et al., 2020], which treats the observed values as control points on a continuous path rather than a discretized sequence – a more literature-faithful match to genuinely irregular ICU sampling than our hourly-binned GRU – and a lightweight self-attention encoder (a standard `nn.TransformerEncoder` [Vaswani et al., 2017], not the specific kernel-attention mechanism of KA-Transformer-style work, which targets inference-time efficiency rather than the accuracy/robustness question this comparison isolates), sized to mirror the GRU’s design (identical input encoding, left-padded/last-position pooling, head shape) so the comparison isolates the encoder itself. Table 8 reports internal AUROC, external AUROC, and the cross-institution gap for both, alongside the GRU baseline.

**Neural CDE: a real trade-off, not a clean win.** The CDE’s external AUROC is significantly *lower* than the GRU’s (Welch’s  $t = -3.10$ ,  $p = 0.023$ ; Cohen’s  $d = -1.96$ ), but its internal-to-external gap is dramatically and significantly *smaller* (0.035 vs. 0.072; Welch’s  $t = -4.82$ ,  $p = 0.003$ ; Mann-Whitney  $p = 0.004$ ; Cohen’s  $d = -3.05$ ), with lower gap variance across seeds too (0.008 vs. 0.015) – exactly the variance reduction the hyperparameter sweep (Section 9.9) failed to achieve, from a different lever.



Table 8: Encoder comparison, 5-seed mean  $\pm$  std. (gap std reported separately for readability). Neural CDE trades peak discrimination for a smaller, more consistent cross-institution gap; attention matches the GRU’s discrimination at under half its parameter count but does not replicate the CDE’s robustness gain.

| Model                           | Params  | Test AUROC        | External AUROC    | Gap          | Gap std      |
|---------------------------------|---------|-------------------|-------------------|--------------|--------------|
| <b>TinySepsis (GRU, ours)</b>   | 191,297 | $0.740 \pm 0.010$ | $0.667 \pm 0.011$ | 0.072        | 0.015        |
| Neural CDE                      | 576,513 | $0.686 \pm 0.006$ | $0.651 \pm 0.005$ | <b>0.035</b> | <b>0.008</b> |
| Attention (Transformer encoder) | 79,681  | $0.745 \pm 0.005$ | $0.663 \pm 0.022$ | 0.082        | 0.023        |

We do not read this as unambiguous evidence that a continuous-time inductive bias improves robustness: a targeted follow-up (one seed, 40 epochs and patience 6 instead of 15/4, isolating training duration as the only changed variable, since per-seed training loss was still declining at epoch 15 for every seed) found validation AUROC did keep improving past epoch 15, but this did *not* translate to better test or external AUROC (test 0.684 vs. 0.693; external 0.647 vs. 0.648, statistically indistinguishable) – more training fits the validation split better without closing the gap to the GRU, consistent with the CDE reaching an effective capacity or inductive-bias ceiling on this task rather than being under-trained. The CDE also costs 576K parameters against the GRU’s 191K and, unlike the GRU and the attention encoder below, does not currently export to ONNX (torchdiffeq’s adaptive solvers do not trace cleanly), a real deployability cost this paper’s own efficiency argument (Section 9) says should be weighed against any accuracy or robustness gain, not set aside.

**Attention: parameter-efficient parity, not a robustness gain.** The attention encoder is statistically indistinguishable from the GRU on both external AUROC (Welch’s  $t = -0.41$ ,  $p = 0.69$ ) and gap (Welch’s  $t = 0.81$ ,  $p = 0.44$ ) at under 42% of the GRU’s parameter count (79,681 vs. 191,297), and – unlike the CDE – exports cleanly to ONNX (numeric parity  $4.8 \times 10^{-7}$  between PyTorch and ONNX outputs, three orders of magnitude tighter than this project’s own  $10^{-3}$  tolerance). Its gap point estimate is numerically a little higher and more variable than the GRU’s ( $0.082 \pm 0.023$  vs.  $0.072 \pm 0.015$ ), not lower, so this is not a repeat of the CDE’s robustness story: it is a credible smaller-footprint candidate if model size becomes a real deployment constraint, with no accuracy cost established on this hospital pair, but not on its own a reason to replace the GRU. Section 9.11 complicates this reading further.

### 9.11 A third, independent external validation: eICU Demo

Every result up to this point compares two institutions from a single data-collection effort (Section 11). We obtained a genuinely independent, if small, third check: the eICU Collaborative Research Database Demo v2.0.1 [Pollard et al., 2018], an openly-licensed, no-credentialing-required 2,520-ICU-stay subsample of eICU spanning **186 distinct hospitals** (per `patient.csv`’s own `hospitalid` field) that overlap none of the Challenge 2019 Hospital A/B institutions this paper otherwise reports on. We built a mapping from eICU’s schema onto our exact 34-feature representation and derived our own simplified Sepsis-3 label (a SOFA-score approximation intersected with antibiotic-order/culture-based suspicion of infection, since eICU does not ship a pre-computed label the way the Challenge 2019 files do; full derivation in `scripts/build_eicu_demo_dataset.py`).

Only **1 of 2,520 patients** meets this simplified Sepsis-3 criterion (6 positive patient-hours total), which we traced to a real data-completeness limit of this specific demo subsample, not a labeling-logic error: eICU’s `microLab` (culture) table has any record at all for only 64/2,520 patients, and only 1 of those also has an antibiotic order in the Sepsis-3 infection window, despite antibiotic detection itself (matched against actual `medication.csv` drug names, not the sparser treatment-category strings) correctly identifying 561/2,520 patients ( $\sim 22\%$ , a clinically plausible ICU antibiotic-exposure rate). The published literature

Table 9: Zero-shot empirical false-positive rate on eICU Demo (2,519 negative-labeled patients, 186 hospitals), at each model’s existing Challenge-2019-calibrated conformal threshold (target  $\alpha = 0.10$ ).

| Model                           | Realized FPR (target 10%) | Alarms/1000h | Per-hospital mean (std) |
|---------------------------------|---------------------------|--------------|-------------------------|
| <b>TinySepsis (GRU, ours)</b>   | <b>0.064</b>              | 63.6         | 0.046 (0.086)           |
| Attention (Transformer encoder) | 0.209                     | 209.3        | 0.166 (0.154)           |
| Neural CDE                      | 0.190                     | 190.2        | 0.168 (0.077)           |

reports  $\sim 4.6\%$  Sepsis-3 prevalence on the *full*, credentialed eICU using the same 6-hour task definition, so we read this as specific to the demo’s culture-documentation sparsity at this small sample size, not evidence that full-eICU replication is infeasible. **This n makes any AUROC or sensitivity claim on this split uninterpretable, and we do not report one.**

What this split *can* test, with its 2,519 negative-labeled patients: whether each model’s own conformal alarm threshold (Section 6, fit only on Challenge 2019 validation data,  $\alpha = 0.10$ ) holds its false-alarm budget on 186 institutions none of the three encoders in Table 8 have ever seen, in training or in calibration. Table 9 reports the result.

The GRU’s realized false-alarm rate (6.4%) stays under its 10% target on hospitals it has never seen, consistent with (if anything more conservative than) its Hospital B behavior (12.2%, Table 4). The attention and CDE encoders do not: both realize an FPR of 19–21%, roughly double their target and triple the GRU’s realized rate here, and the elevation is broad across institutions rather than driven by a few outlier hospitals (per-hospital mean alarm rate 16.6% and 16.8–18.1% respectively, vs. the GRU’s 4.6%). **We read this as a direct caution about Section 9.10’s own comparisons:** both alternative encoders looked statistically competitive with (attention) or offered a genuine, documented trade-off against (CDE) the GRU when judged on the Hospital A/B pair alone, but on a third, previously unseen set of institutions, their calibrated decision thresholds generalize substantially worse than the GRU’s – the same ranking-transfers/threshold-transfers-worse asymmetry Section 10 already documents for TinySepsis itself against the tabular baselines, now recurring one level down, between TinySepsis and its own architecture variants. Two institutions were not enough to catch this; we would not have known it without a third. This does not settle which encoder is best – eICU Demo’s own label sparsity means we cannot check whether the false-alarm elevation trades off against better sensitivity, the one measurement this split cannot make – but it is concrete evidence against adopting either alternative encoder on the strength of the Hospital A/B comparison alone, and the clearest illustration in this paper of why we treat two-institution external validation as a floor, not a sufficient standard.

## 9.12 Multi-horizon auxiliary training

Sections 9.9–9.11 exhaust the levers this paper originally set out to check; we tried two further ones that use *zero new data* – only the labels and checkpoints already produced above – specifically because they were untested and because a natural question after Section 9.11 is whether anything short of a new institution can still move the external number. The first: TinySepsis predicts a single 6-hour horizon, but `label_4h` and `label_8h` are already computed for every row (Section 3) and share almost all of their signal with `label_6h`, differing only in how close to  $t_{\text{susp}}$  the positive window starts. We widened the final layer to predict all three jointly (a trivial parameter cost: 191,427 vs. 191,297) and trained on their sum, evaluating, as everywhere else in this paper, only the primary 6h task. Table 10 reports the result on the same 5-seed protocol as Table 3.

Internal AUROC is higher with substantially lower seed-to-seed variance (std 0.004 vs. 0.010), a real regularization signal. External AUROC is also higher on average (0.679 vs. 0.667), but the two significance tests we report throughout this paper disagree at the conventional threshold here: Welch’s  $t = 1.73$ ,  $p =$

Table 10: Multi-horizon (joint 4h/6h/8h) auxiliary training vs. single-horizon TinySepsis, 5-seed mean  $\pm$  std, evaluated on the primary 6h task only.

| Model                                 | Test AUROC                          | External AUROC                      | Gap               |
|---------------------------------------|-------------------------------------|-------------------------------------|-------------------|
| TinySepsis (GRU, single-horizon)      | $0.740 \pm 0.010$                   | $0.667 \pm 0.011$                   | $0.072 \pm 0.015$ |
| <b>Multi-horizon (4h+6h+8h joint)</b> | <b><math>0.750 \pm 0.004</math></b> | <b><math>0.679 \pm 0.010</math></b> | $0.070 \pm 0.011$ |

Table 11: Probability-averaging ensembles of already-trained checkpoints, no retraining. Each row adds the previous row’s models plus one more architecture or seed group.

| Ensemble                                  | Test AUROC | External AUROC |
|-------------------------------------------|------------|----------------|
| Single GRU seed (reference)               | 0.726      | 0.679          |
| GRU, 5 seeds averaged                     | 0.755      | 0.685          |
| GRU + Attention, 10 models averaged       | 0.763      | 0.685          |
| GRU + Attention + CDE, 15 models averaged | 0.770      | <b>0.702</b>   |
| + multi-horizon seeds, 20 models averaged | 0.771      | 0.702          |

0.121 (not significant) against Mann-Whitney  $p = 0.048$  (marginally significant), Cohen’s  $d = 1.10$  (large). We report this honestly as a borderline result, neither the clean null of Section 9.9 nor the unambiguous significance of the ensembling result below – and, given it costs nothing to keep, a reasonable candidate default for future training runs of this model regardless of whether the external-AUROC gain replicates further.

### 9.13 Ensembling across seeds and architectures

The second zero-new-data lever: simple probability-averaging across models already sitting on disk from Sections 9.3, 9.10, and 9.12 above – no retraining. Significance here is via row-level bootstrap (300 resamples) rather than the seed-based tests used elsewhere, since an ensemble of 5 seeds is one deterministic aggregate, not 5 independent draws; row-level (not patient-clustered) resampling ignores within-patient correlation and likely understates the true confidence interval width, so we read this evidence as suggestive, not at the same standard as the rest of this paper. Table 11 reports AUROC as models are added.

Every addition through the 15-model (GRU + attention + CDE, 5 seeds each) ensemble is bootstrap-positive in 100% of resamples: averaging the GRU’s own 5 seeds beats a single seed (external  $+0.0056$ , 95% CI  $[0.0024, 0.0090]$ ), and – the more novel finding – averaging across *architectures*, not just seeds, adds real additional value on top of that (external  $+0.0173$  further, CI  $[0.0148, 0.0197]$ ). This is, to our knowledge, not previously reported for this task: an encoder ensemble outperforms any single encoder or same-architecture seed ensemble on external AUROC, including the CDE and attention variants whose individual comparisons in Sections 9.10–9.11 were mixed or unfavorable relative to the GRU alone. Adding the multi-horizon seeds on top gets a negligible further  $+0.0004$ , and only 80% of bootstrap resamples agree it is even positive – diminishing returns have set in by 15–20 models. **The best external AUROC obtained anywhere in this paper, 0.702, combines every model trained across both rounds of this analysis into one 15-model ensemble**, a genuine  $+3.5$  percentage-point improvement over the single-model 5-seed mean (0.667), for the cost of running inference through 15 checkpoints instead of one and no additional training.

We did not refit calibration for any ensemble. Every number in Table 11 is raw AUROC on averaged probabilities; a deployed ensemble would need its own temperature/isotonic/conformal-threshold fit (Section 6) before the utility and false-alarm-rate claims made elsewhere in this paper could be trusted for it – an

improved AUROC does not imply the averaged probabilities are well-calibrated, and we have not checked whether it is.

## 10 Discussion

The central empirical question of this paper is not "does TinySepsis beat XGBoost on AUROC" — with under 200K parameters and no hyperparameter search, we did not expect it to dominate a well-tuned gradient-boosted tree on a tabular-friendly task, and on the internal test set it does not, in any of the 5 seeds we trained (Table 1, Table 3: XGBoost  $0.805 \pm 0.003$  vs. TinySepsis  $0.740 \pm 0.010$  mean AUROC). The result we consider load-bearing is the *gap*: every tabular baseline loses 0.115–0.126 AUROC on average moving from Hospital A to Hospital B, while TinySepsis loses only 0.072, and Section 9.3 shows this is not a lucky seed — TinySepsis’s degradation was smaller than all 15 baseline-seed runs, with  $p \leq 0.002$  (Welch) and  $p = 0.004$  (Mann-Whitney) against each baseline after Bonferroni correction. On the institution none of the models ever saw during development, TinySepsis trails XGBoost by a modest margin on average (mean external AUROC 0.667 vs. 0.686) and outperforms logistic regression (0.644) despite starting well behind both internally. We did not tune any model to produce this pattern — baselines used default-adjacent hyperparameters and the same feature set as TinySepsis (Section 8) — so the most parsimonious reading is that the higher-capacity tabular models are fitting institution-specific structure (Hospital A’s particular case mix, documentation habits, or equipment) that a smaller, more constrained recurrent model does not have the capacity to overfit to, at the cost of leaving internal AUROC on the table. This is exactly the failure mode reported for the Epic Sepsis Model in external deployment [Wong et al., 2021, Habib et al., 2021], reproduced here in miniature and under our control. We treat it as the paper’s main finding, not a secondary robustness footnote; it is now supported by more than a single run, but it remains evidence from a single hospital pair, and we do not know a priori whether it will replicate on a third institution (Section 11) — that replication is the most direct follow-up we can name. Two further comparisons matter beyond this headline result: (1) whether calibration and conformal risk control meaningfully change the alarms-per-1000-hours and normalized-utility numbers at a fixed sensitivity target, since those are the numbers that determine whether clinical staff tolerate a system long enough for it to matter [Sendak et al., 2020] — here isotonic regression collapses ECE by roughly two orders of magnitude (Table 4) while the conformal alarm-rate budget (target  $\alpha = 0.10$ ) holds reasonably on the internal split but visibly inflates on Hospital B, a second, independent demonstration of the same cross-institution fragility; and (2) whether the missingness ablation (Table 7) shows that explicitly modeling *what wasn’t measured* carries real signal beyond the measured values themselves — removing the mask/time-since-last-measurement channels costs 0.037 AUROC internally, consistent with informative missingness being a first-class modeling concern in ICU time series rather than an implementation detail [Che et al., 2018], though its effect on external AUROC in our single-run comparison was noisier and should not be over-read (Section 11).

The headline finding needs one more qualification we did not anticipate when we designed the experiment: it holds for *ranking* (AUROC) but not for *calibrated decision thresholds*. Table 5 shows that TinySepsis’s own conformal threshold, fit on Hospital-A validation data exactly as prescribed in Section 6, controls the false-alarm rate almost exactly to the 10% target internally but rises to 12.2% on Hospital B — roughly 2–3 $\times$  the false-alarm rate XGBoost or LightGBM realize on the same external split with their own conformal thresholds. The net effect is that TinySepsis is the only learned model whose normalized utility goes *negative* on Hospital B at its recommended operating point (−0.023, essentially break-even), while the three tabular baselines all stay comfortably utility-positive there (+0.12 to +0.16). So the paper’s two central claims point in opposite directions depending on which axis of robustness is asked about: TinySepsis’s *relative ranking* of patients transfers across institutions better than the tabular baselines’, but its *absolute calibrated risk estimate* — the number a clinician would actually see, and the one the alarm threshold is set

against – transfers worse. We do not have a confident explanation for this asymmetry; a plausible one is that a single global temperature/isotonic map is less able to correct a small recurrent model’s shift in output distribution under a change in base rate (0.9% vs. 1.4% prevalence) than it is for tree-ensemble scores, but we did not design an experiment to isolate this and flag it as the second highest-priority follow-up after the multi-seed replication already discussed above. A reader taking one number away from this paper should not take "TinySepsis is the robust choice" – they should take "AUROC robustness and calibrated-threshold robustness are different properties that must both be measured, because a model can have one without the other."

Sections 9.9–9.13 tried five separate levers to move past this picture and are worth reading together rather than as five independent results. Hyperparameter tuning (Section 9.9) does not reduce TinySepsis’s seed-to-seed variance or improve its external AUROC at statistical significance – the variance is not a regularization problem. Architecture change (Section 9.10) does move the needle, but along two different axes that do not both improve together: the Neural CDE trades lower absolute discrimination for a significantly smaller and less variable cross-institution gap, while attention matches the GRU’s discrimination at under half its parameters without inheriting the CDE’s robustness gain. The eICU Demo check (Section 9.11), despite being too small to say anything about sensitivity, was large and diverse enough (186 hospitals) to reveal that both alternative encoders’ calibrated alarm thresholds generalize substantially worse than the GRU’s to institutions outside the Hospital A/B pair – a result neither the hyperparameter sweep nor the two-hospital architecture comparison could have surfaced. Multi-horizon training (Section 9.12) gave a real but statistically borderline external-AUROC gain at zero cost. And ensembling everything trained across this whole analysis (Section 9.13) gave the single largest, most confidently significant external-AUROC improvement of any lever tried – 0.702, up from 0.667 – for the cost of running inference through 15 checkpoints rather than one, with the added, not-previously-obvious finding that averaging across *architectures* helps more than averaging across seeds of one architecture alone. Put plainly: every axis we checked this round, checking mattered more than the specific technique – the sweep would have been misread as a win at 3-seed screening, the CDE would have been read as a clean improvement without the external-AUROC significance test, attention would have been read as a safe drop-in replacement without a third institution to test it against, and the ensemble’s real gain would have been missed entirely without trying it at all. We take this as reinforcing, not just repeating, the paper’s central methodological argument. We also do not think 0.702 is close to a ceiling this project could push toward 0.90 external AUROC by further recombination of the same techniques: every lever that touches the model or training recipe rather than the training data’s institutional diversity has now been tried on this hospital pair, and the pattern – null, modest-and-borderline, or a real trade-off, never large – across all of them points at a limit set by the data available, not by remaining technique.

## 11 Limitations

- **Single hospital pair, still.** All data originate from the PhysioNet Challenge 2019 release. Hospital A vs. Hospital B is a genuine, uncurated distribution shift, but it is one pair of institutions. Section 9.3 addresses *random-seed* variance directly: we retrained every learned model under 5 seeds and found TinySepsis’s internal-to-external AUROC gap smaller than all 15 baseline-seed runs, with  $p \leq 0.002$  (Welch) and  $p = 0.004$  (Mann-Whitney) against each baseline after Bonferroni correction, so the headline finding is not an artifact of a favorably-chosen initialization. What this does *not* address is *hospital-pair* variance: all 5 seeds share the same Hospital-A-train/Hospital-B-test split, so a systematic quirk of this specific pair of institutions (rather than of "internal vs. external" in general) would inflate every seed identically and our test would not catch it. We also note that logistic regression’s solver (`lbfgs`) is deterministic given fixed data, so its "5 seeds" are 5 copies of one run, not 5 independent draws – we still count it as  $n = 5$  in



the statistical tests above for consistency with the other baselines, which is conservative in TinySepsis’s favor only in the trivial sense that a zero-variance comparator cannot narrow the reported confidence; the effect sizes against XGBoost and LightGBM, which do have real seed-to-seed variance, are the ones that matter most. None of this is the same as validating against the full, credentialed MIMIC-IV or eICU databases, which use different monitoring equipment, EHR systems, and documentation practices [Johnson et al., 2023, Pollard et al., 2018]; we did not have credentialed access to those within the scope of this study (obtaining it requires PhysioNet’s CITI training and a signed, individual, non-transferable Data Use Agreement, a human process outside this paper’s automatable pipeline, and its license would in any case prohibit redistributing a model trained on it through this paper’s public code release without a separate access-controlled distribution channel). We note in passing that MIMIC-IV is single-center (Beth Israel Deaconess Medical Center only) and so would not by itself add cross-institution diversity the way eICU’s 200+-hospital design would. We did obtain the openly-licensed eICU Demo subsample (Section 9.11), which is genuinely useful – 186 institutions, no credentialing barrier – but too small and too sparse in its microbiology-culture documentation (Section 9.11) to support a sensitivity or AUROC claim; it answers a false-alarm-generalization question, not a detection one. A full MIMIC-IV/eICU replication – ideally with multiple hospital pairs, not just one more single pair, and with a properly adjudicated rather than approximated Sepsis-3 label – remains the single highest-priority follow-up.

- **Retrospective, not prospective.** All evaluation is retrospective on already-collected, already-labeled data. Retrospective AUROC and even retrospective alarm rates do not guarantee that a deployed system changes clinician behavior or patient outcomes; only a prospective, ideally randomized, evaluation can establish that, and none was performed here.
- **Approximate clinical scores.** Our qSOFA and NEWS2 baselines are necessarily incomplete (no Glasgow Coma Scale, no supplemental-oxygen flag in the source data), which likely understates their true achievable performance and should not be read as a definitive comparison against full-criteria clinical scores.
- **Label definition sensitivity.** We reconstruct  $t_{\text{susp}}$  from the Challenge’s own labeling convention rather than re-deriving Sepsis-3 criteria from raw data; sensitivity of our results to alternative sepsis-onset definitions is not evaluated here.
- **The eICU Demo label is a simplified approximation, not a clinically adjudicated one.** Section 9.11’s SOFA score omits or approximates several official components (mechanical-ventilation status for the respiratory score, cumulative urine output for the renal score, continuous vasopressor dosing collapsed to eICU’s own bucketed dose-tier strings for the cardiovascular score) and treats a missing component as 0 rather than imputing it, a conservative choice that likely under-, not over-, counts true organ dysfunction. Combined with that split’s near-total absence of positive labels, this is sufficient for the false-alarm-generalization check we report and not for anything else; we did not and could not validate it against chart-reviewed ground truth.
- **Conformal coverage under shift.** The conformal risk control guarantee is a marginal, exchangeability-dependent guarantee; Table 4 reports, rather than hides, its degradation on the external split, and that degradation should be treated as expected and diagnostic, not as a bug.
- **Demographic granularity.** The dataset provides only age and a binary sex field; no race, ethnicity, or insurance-status fields are available for a fuller equity audit, which limits the subgroup analysis in Section 9 to two axes.

## 12 Ethical Considerations

TinySepsis is a research artifact. It is **not** a validated medical device, has not undergone FDA clearance, CE marking, or any equivalent regulatory review, and must not be used to inform real clinical decisions. All underlying data are drawn from a fully de-identified, publicly released research dataset governed by PhysioNet’s open data license; no new patient data were collected, and no protected health information is processed by any code in this repository. The accompanying local demo (Section 13) displays an explicit "not for clinical use" disclaimer on every response and is intended solely to demonstrate that inference can run fully offline, on commodity hardware, without transmitting any data to a third party. Given the Epic Sepsis Model precedent [Wong et al., 2021, Habib et al., 2021], we consider it an ethical obligation — not merely good scientific practice — to report external-hospital degradation prominently rather than to lead with the more favorable internal numbers, and we have structured Section 9 accordingly. Any future clinical translation of this or a similar system would require, at minimum: prospective validation at the deployment site, a locally-fit conformal calibration (not a transplanted one), a human-in-the-loop workflow with an explicit override path, and institutional review board oversight.

## 13 Local Deployment and Reproducibility

The trained model is exported to ONNX (opset 17) and served through a minimal FastAPI application (`tinysepsis.demo.app`) that accepts hourly observations and returns a calibrated risk probability, a conformal alarm decision, and the top contributing feature channels via Integrated Gradients [Sundararajan et al., 2017], computed from the PyTorch checkpoint the deployed ONNX graph was exported from (verified numerically equivalent at export time) purely for this explanation step, while ONNX continues to serve the score itself. An earlier version of this field used  $|z\text{-score}|$  of the last observed raw value as a proxy for importance – a measure of how unusual a value looked, not of what the model actually responded to, which could rank a feature the model largely ignored above one driving the prediction. We replaced it after noticing this is precisely the kind of gap the Cures Act non-device clinical-decision-support criterion of “enabling the healthcare professional to independently review the basis for the recommendation” is meant to rule out: the basis shown has to be the model’s actual basis. Correctness of the new attribution is enforced by testing Integrated Gradients’ completeness axiom (attributions must sum to  $f(x) - f(x')$  for baseline  $x'$ ) rather than by inspection. The demo runs entirely on CPU with no external network calls. The complete pipeline — dataset download, feature engineering, training, baseline training, calibration, conformal fitting, ONNX export, and evaluation — is implemented as a sequence of standalone scripts (`scripts/*.py`) intended to be run in order; exact commands, environment specification, and hardware requirements are documented in the repository `README.md`. All randomness-affecting steps use a fixed seed (Section 4) for patient-level splitting; full bitwise reproducibility of GPU-trained floating-point results is not claimed, consistent with known non-determinism in cuDNN/cuBLAS kernels.

## 14 Conclusion

We presented TinySepsis, a sub-200K-parameter recurrent model for early sepsis prediction designed around three properties we argue are under-weighted relative to raw discrimination in the sepsis-prediction literature: calibration, an explicit and auditable false-alarm budget via conformal risk control, and measured (not asserted) deployability on commodity hardware. Trained and evaluated entirely on the open-access PhysioNet Challenge 2019 dataset, with a cross-institution evaluation standing in – imperfectly, as discussed in Limitations – for the kind of independent-cohort evaluation whose absence contributed to a well-documented real-world failure [Wong et al., 2021], TinySepsis is offered as a fully reproducible baseline and a method-



ological argument: for clinical early-warning systems, how a model fails under distribution shift and how honestly its uncertainty is represented are not secondary concerns to be mentioned in a limitations paragraph, but primary evaluation criteria that belong in the main results table. Our own results make the case for that argument in a way we did not plan: TinySepsis’s ranking of patients (AUROC) generalized across hospitals substantially better than the tabular baselines’, but its calibrated alarm threshold generalized worse, and only measuring the first property would have produced a paper with the opposite conclusion from measuring both. We take that as the strongest evidence in this paper that discrimination alone is an insufficient basis for judging a clinical early-warning system – a conclusion that holds up under the 5-seed replication in Section 9.3 ( $p \leq 0.002$  against every tabular baseline), though it is still evidence from one hospital pair, and a multi-institution replication remains the necessary next step (Section 11).

A post-hoc round of model-improvement experiments (Sections 9.9–9.13) reinforced this argument in a direction we did not plan for either: neither hyperparameter tuning nor two credible architecture alternatives (a Neural CDE, a lightweight attention encoder) produced an unambiguous improvement once checked at the same standard of evidence – multiple seeds, significance tests, and, this time, a third external institution beyond the Hospital A/B pair. That third check was the one that mattered most: two encoders that looked competitive with or better than the GRU on Hospital B alone both showed substantially worse false-alarm generalization on 186 hospitals neither had ever seen, a failure mode invisible to the two-institution comparison this paper otherwise relies on throughout. We did not adopt either alternative encoder as a result, and see that decision – not any single number in Table 8 or 9 – as this addendum’s central contribution: checking claimed improvements against genuinely new institutions, not just genuinely new seeds, is not optional due diligence but a distinct axis of evidence a model can fail on its own.

Two further zero-new-data levers – multi-horizon auxiliary training and ensembling across every model trained in this analysis – gave the round’s only real gains, and the second gave by far the largest: 0.702 external AUROC, the best number anywhere in this paper, from simply averaging 15 already-trained checkpoints spanning three architectures. We report it with the same qualification we apply to everything else here: it is not calibrated (Section 9.13), it is evaluated with a weaker bootstrap significance standard than the seed-based tests elsewhere, and, more fundamentally, 0.702 is still far from a number we would call solved. We now believe, on the strength of having tried and reported every technique-side lever available on this hospital pair, that closing the remaining distance requires institutional diversity this dataset cannot provide, not further recombination of the techniques in this paper – a belief we hold with more confidence for having spent this addendum’s effort disproving it, on this dataset, at every turn we could find.

## Acknowledgments

This work uses the PhysioNet/Computing in Cardiology Challenge 2019 dataset [Reyna et al., 2019b] and the openly-licensed eICU Collaborative Research Database Demo [Pollard et al., 2018], made available by their respective organizers and contributors; we thank them for enabling fully reproducible research on early sepsis prediction without a credentialing barrier.

## Code and Data Availability

Code, trained model weights, the exact feature-engineering pipeline, and scripts to regenerate every table and figure in this paper from the raw, publicly downloadable dataset are available at <https://github.com/elianalfonsolopezpreciado/tinysepsis> (MIT license). The underlying clinical dataset is not redistributed; `scripts/download_data.py` fetches it directly from PhysioNet’s open-access S3 mirror.

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